

1	(a)	The rate of change of momentum of an object <u>is directly proportional to the resultant force</u> acting on the object and has the <u>same direction as the force</u> .		B1 B1
	(b)	(i)	Let a be the magnitude of the acceleration for both block X and block Y. Resultant force on system = gravitational force pulling block X along the ramp - gravitational force pulling block Y downwards $(3.0 + 1.2) a = 3 \times 9.81 \times \sin 30^\circ - 1.2 \times 9.81$ $a = 0.70071 \approx 0.701 \text{ m s}^{-2}$	C1 A1
		(ii)	$v^2 = u^2 + 2as$ $v^2 = 0 + 2(0.70071)(0.5)$ $v = 0.83708$ $= 0.837 \text{ m s}^{-1}$	C1 A1
		(iii)	Let the take-off speed at the ramp be v'. Considering the next 0.5 m moved by block X, we have $v'^2 = v^2 + 2as$ $v'^2 = 0.83708^2 + (2 \times 9.81 \sin 30^\circ \times 0.50)$ $v' = 2.3676 \approx 2.37 \text{ m s}^{-1}$	M1 A1
		(iv)	Considering the projectile motion, and taking downwards as positive, $s_y = v'_y t + \frac{1}{2} a t^2$ $1.5 = (2.3676 \sin 30^\circ) t + (\frac{1}{2} \times 9.81 \times t^2)$ $4.905 t^2 + 1.1838 t - 1.5 = 0$ $t = 0.44534 \text{ s}$ $s_x = v'_x t$ $s_x = 2.3676 \cos 30^\circ \times 0.44534$ $s_x = 0.91313 \text{ m}$ Total displacement $= 0.91313 + 1.0 \cos 30^\circ$ $= 1.7792 \approx 1.78 \text{ m}$	C1 C1 M1 A1